Entity Relationship Diagram

Supermarket Management System

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Entity-Relationship Diagram

The following section outlines the Entity-Relationship (ER) model designed for the Supermarket Management System. This model serves as the blueprint for the system's underlying database, detailing how crucial pieces of information are structured, stored, and interconnected.

The database architecture comprises four primary entities, each with specific attributes necessary for day-to-day operations:

- **Customer:** Represents the individuals purchasing goods. Key attributes include a unique system identifier (`Cust_ID`) and the customer's full name (`Cust_Name`).
- **Order:** Represents a transaction made by a customer at checkout. It is tracked using a unique order identifier (`Ord_ID`) and the exact date the transaction occurred (`Ord_Date`).
- **Product:** Represents the physical inventory available for sale. It encompasses a unique product code (`Prod_ID`), the descriptive name of the item (`Prod_Name`), and its current retail price (`Price`).
- **Employee:** Represents the staff members operating the supermarket. Attributes include a unique staff identifier (`Emp_ID`) and their specific job designation or role (`Emp_Role`).

To accurately reflect the real-world business logic of the supermarket, the ER diagram also maps out the following relationships and their cardinalities:

- **Places (1:N):** A single *Customer* can place multiple *Orders* over time. This one-to-many relationship ensures that customer purchase histories can be accurately maintained for potential loyalty programs or return tracking.
- **Contains (M:N):** An *Order* can contain numerous *Products*, and simultaneously, a specific *Product* can appear in many different *Orders*. This many-to-many relationship captures the standard behavior of items in a shopping cart.
- **Manages (1:N):** A single *Employee* is responsible for processing or managing multiple *Orders*. This one-to-many relationship establishes clear accountability for every transaction processed within the system.

A full-page, landscape-oriented view of the resulting diagram, visually depicting these entities, attributes, and relationships, can be found on the following page.

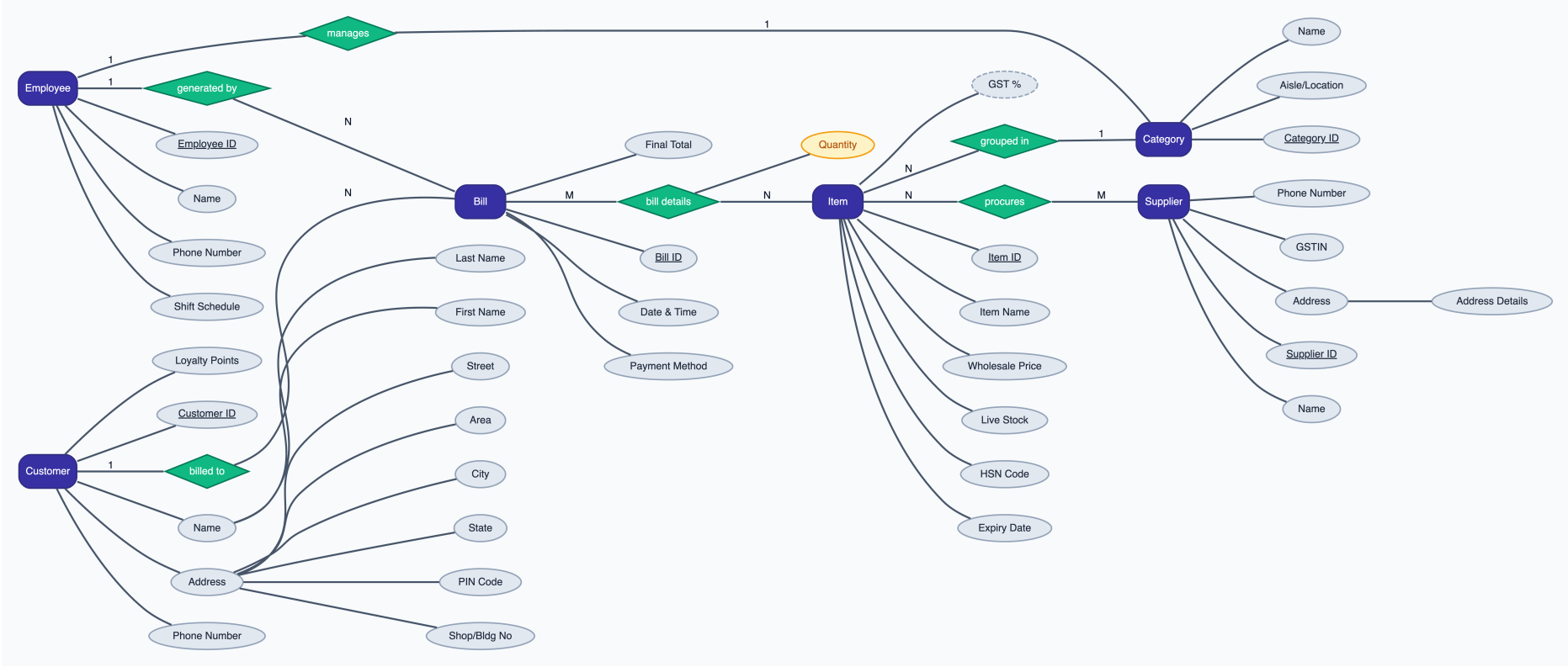


Figure 1: Entity-Relationship Diagram for the Supermarket Management System